

Mapping Inequality on the Boston Plate

Neighbourhood Socioeconomics & Food Inspection Outcomes in Boston, 2015–2025

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RESEARCH QUESTION

Do food safety inspection outcomes in Boston systematically differ across neighbourhoods of varying socioeconomic status — and has any such gap widened over the past decade?

MAIN RESULT

Yes, but the effect is small — and it has not widened. Establishments in low-income tracts have a ~ 3.4 percentage-point lower predicted pass probability than those in high-income tracts, and this gap has remained statistically stable across the 2015–2025 window. Most variation in pass/fail is within-establishment noise rather than neighbourhood-level disparity.

i. Background & Motivation

Public food inspection records document day-to-day enforcement of health regulations across an entire city. If neighbourhood disadvantage shapes whether establishments pass or fail, it would suggest disparities in regulatory protection across communities. Boston offers a ten-year, geocoded inspection record linked to census tract demographics, giving us a natural test case for asking whether neighbourhood SES predicts inspection outcomes after accounting for establishment-level differences.

ii. Data Sources

Ten years of Boston inspections linked to tract-level Census demographics.

Boston Food Establishment Inspections (2015–2025) — 872,111 raw violation records, deduplicated to 100,211 unique inspections at the establishment-visit level. **ACS 2019 5-year estimates** at census-tract level for income, poverty, racial composition, and foreign-born share. Spatially joined to inspections via tract polygons.

The overall 46% fail rate reflects Boston’s inclusive definition of *fail* — any inspection with at least one violation of any severity. Most failed inspections involve minor, correctable issues rather than public health emergencies.

91.6K

INSPECTIONS

6,912

ESTABLISHMENTS

174

CENSUS TRACTS

46%

OVERALL FAIL RATE

iii. Methods

- **Mixed-effects logistic regression** via `glmmTMB` with random intercepts by establishment.
- **Sequential model building**: null $\rightarrow$ income-only $\rightarrow$ SES composite $\rightarrow$ fully adjusted $\rightarrow$ income $\times$ time interaction.
- **PCA-based SES index** to address collinearity ($r = -0.81$ income–poverty). PC1 captures 79%.
- **Train/test split** by establishment (80/20) to avoid leakage.
- **Robustness**: COVID exclusion, nonwhite check, DHARMA diagnostics.
- **Model diagnostics**: residual analysis via DHARMA, AUC evaluation on held-out establishments.

★ KEY FINDING

A 1 SD increase in tract income ($\sim \$40K$) raises the odds of passing by **6.4%**.

A low-income Food Service establishment has a **50.1%** predicted pass probability, vs **53.5%** in a high-income tract. The gap is real but small.

iv. Exploratory Findings

At the tract level, the bivariate relationship between fail rates and median income is **weakly negative but visually flat**, which implies that any neighbourhood SES effect will be modest.

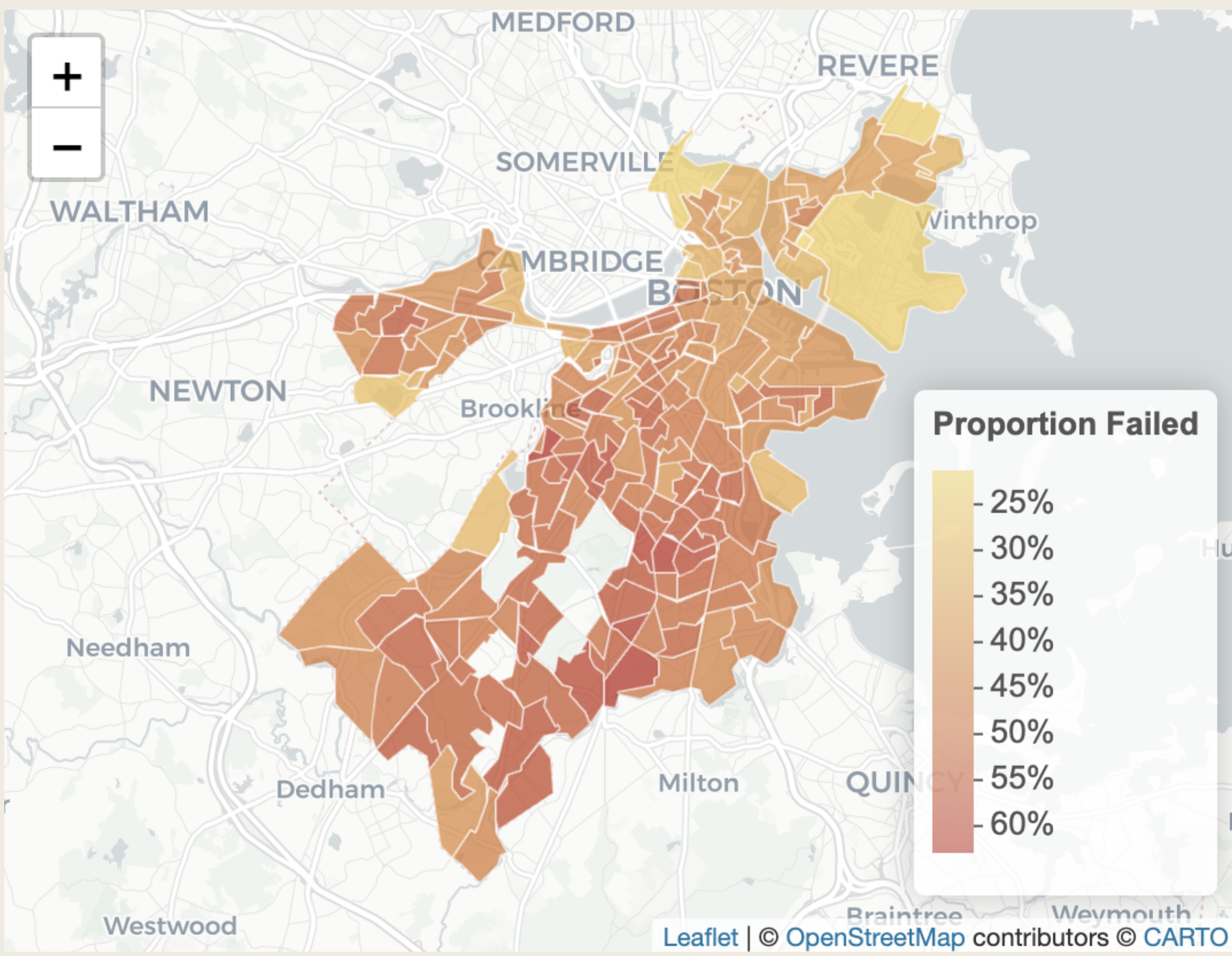


Figure 1. Inspection fail rates by Boston census tract. Darker tracts indicate higher proportions of failed inspections; the spatial pattern motivates the neighbourhood-level analysis.

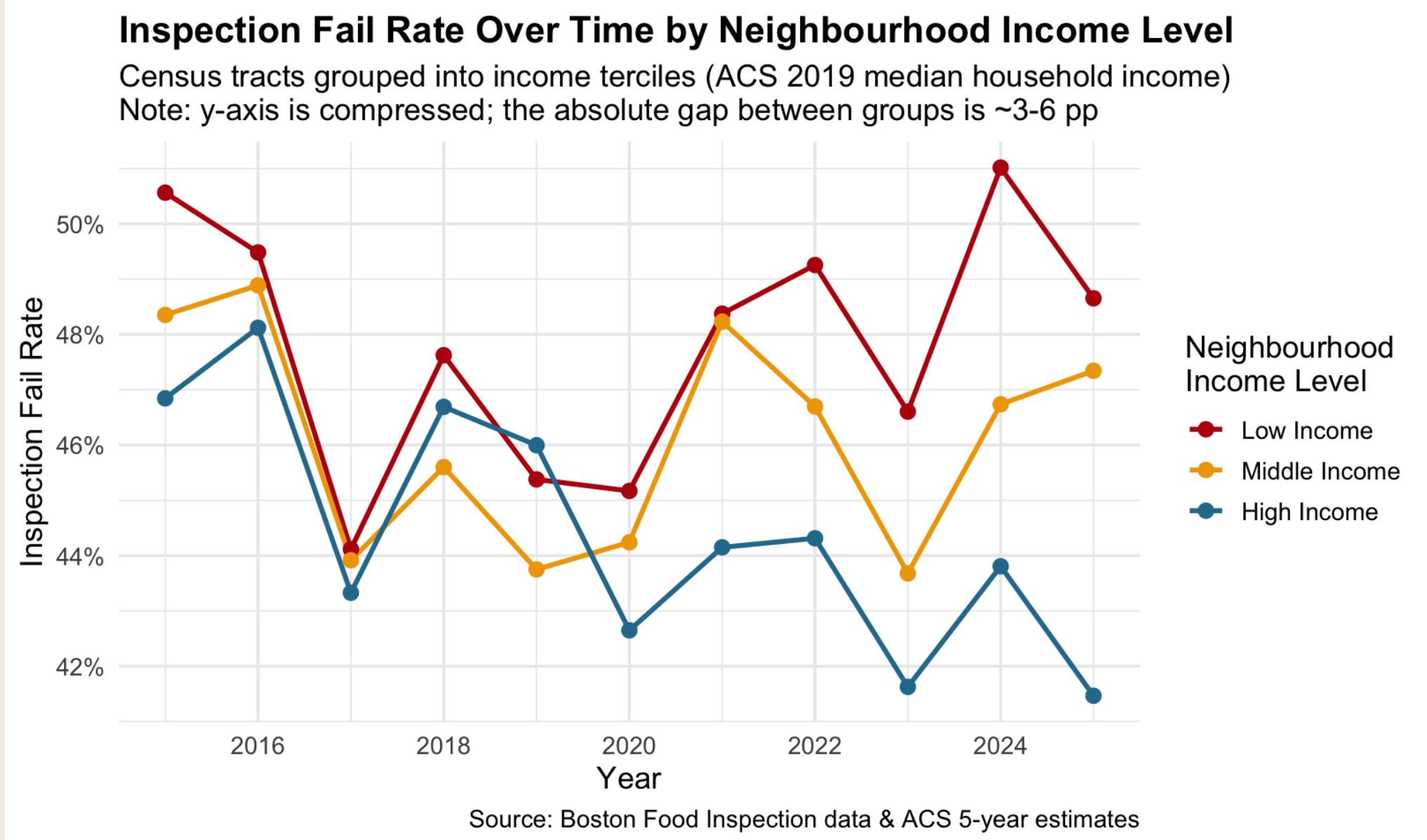


Figure 2. Tract-level fail rates show only a weak negative slope across the income distribution. Each bubble is a census tract sized by inspection volume.

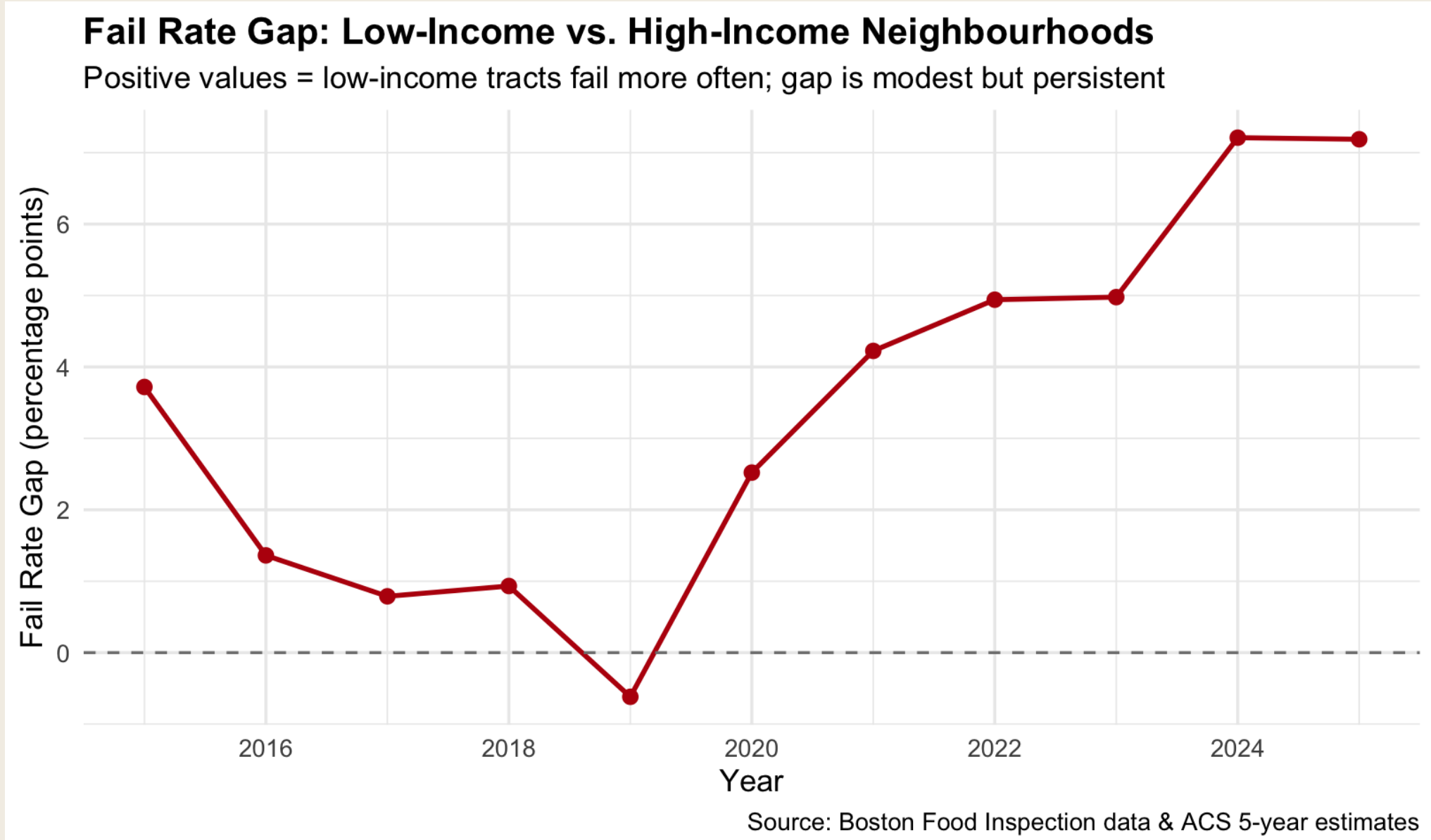


Figure 3. Low-income tracts consistently fail more often than high-income ones, but the absolute gap is only ~ 3 –6 pp on a compressed y-axis.

v. Modelling Results

The **null model ICC is just 0.046** — only 4.6% of pass/fail variation lies between establishments. The remaining 95% is within-establishment variation from visit to visit, suggesting that inspection outcomes are dominated by visit-level factors (inspector, day, conditions) rather than stable establishment characteristics. This places a hard ceiling on how much any neighbourhood-level predictor can explain. Adding fixed effects further shrinks the random-effect variance by **87.7%**, leaving very little residual establishment heterogeneity.

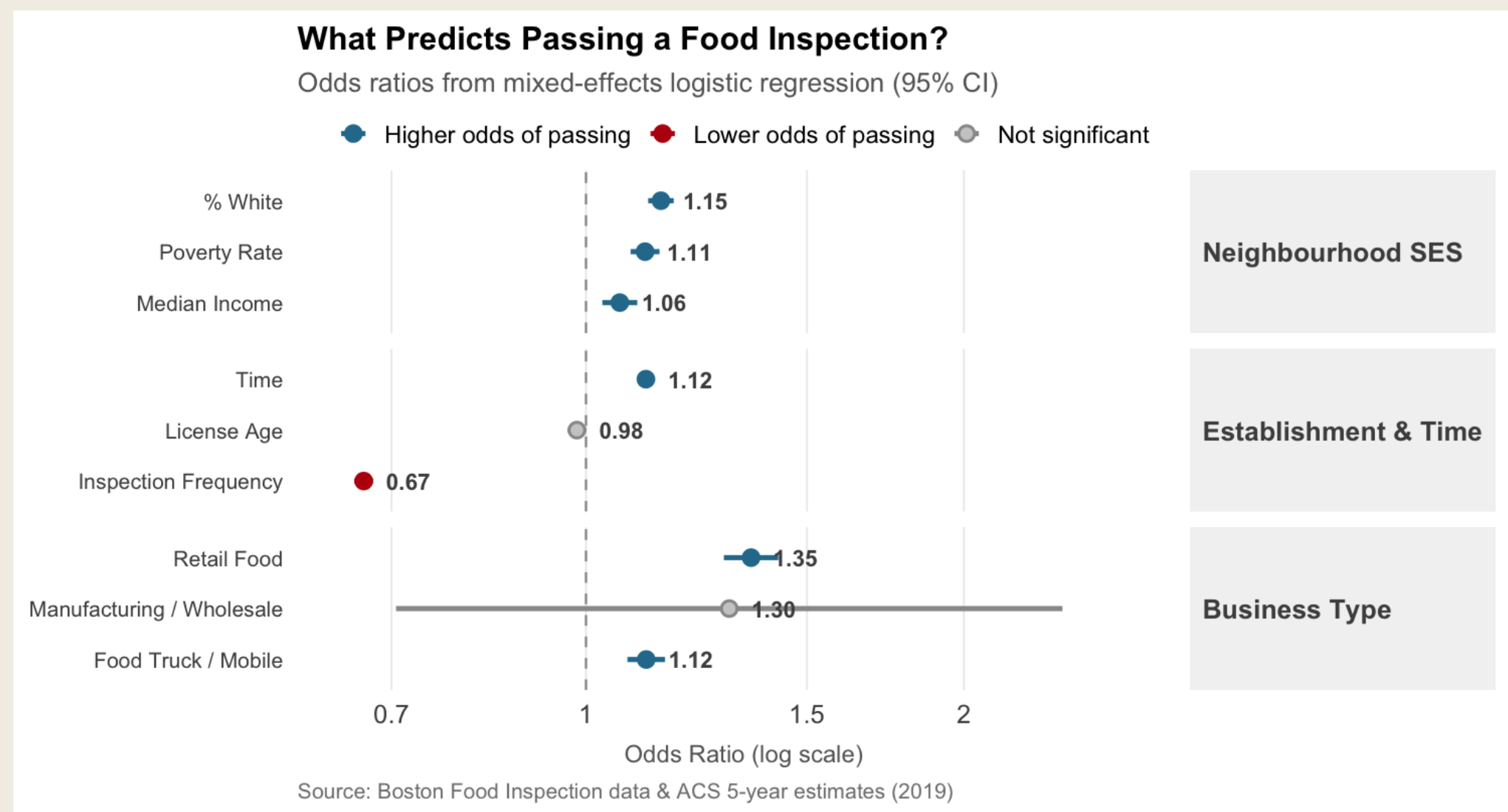


Figure 4. Inspection frequency dominates the coefficient plot ($OR \approx 0.67$). Median income is positive but modest ($OR \approx 1.06$).

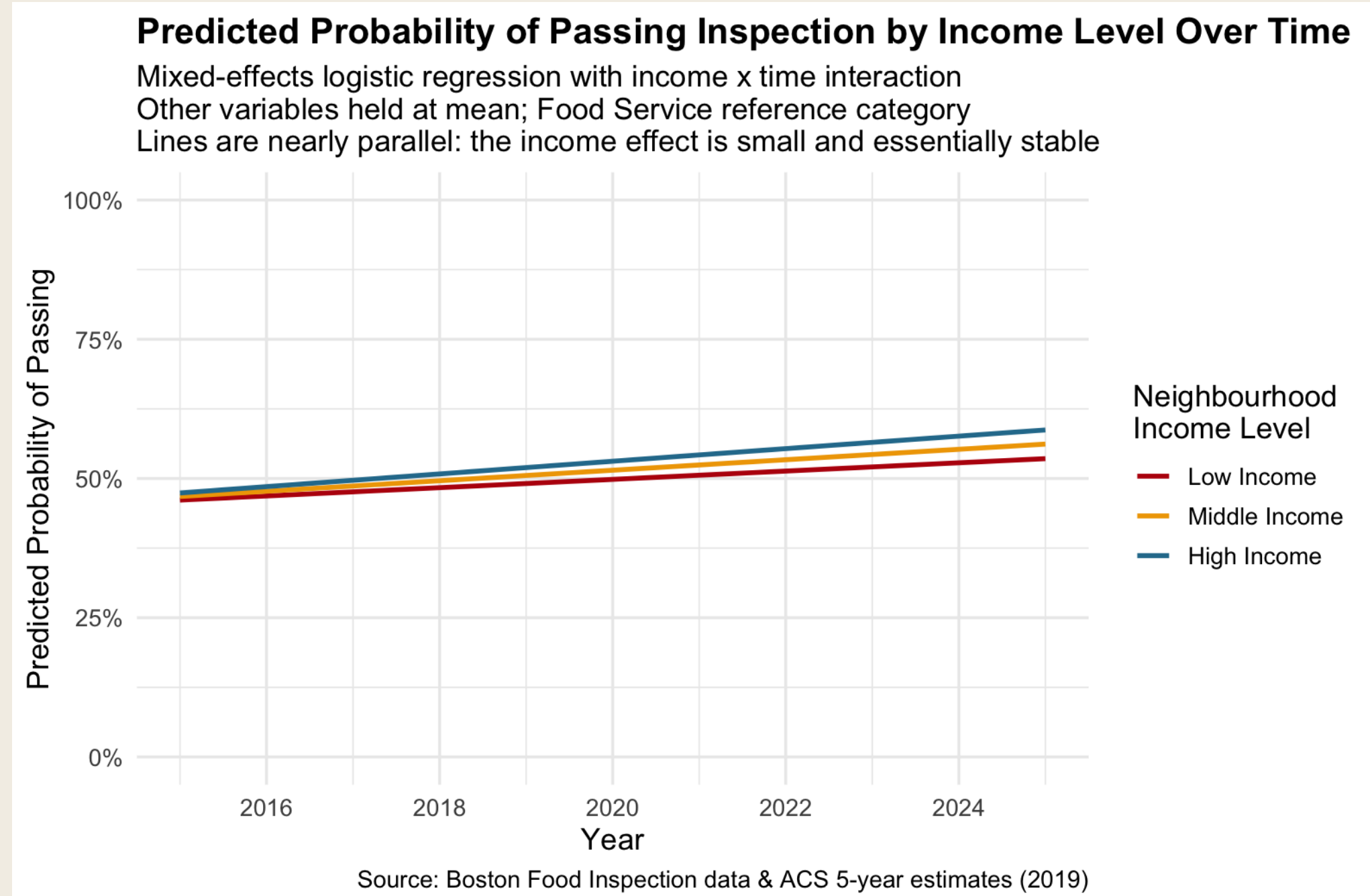


Figure 5. Predicted-probability lines for low/middle/high income tracts are nearly parallel — the income $\times$ time interaction is significant ($p = 0.006$) but practically negligible ($OR = 1.022$).

Limitations

Single-city analysis. ACS 2019 demographics applied to a 10-year window. Pass/fail is coarse and may not capture violation severity. Inspection frequency is endogenous to outcomes. Spatial autocorrelation not modelled. COVID sensitivity check showed all ORs shift < 0.02 ; $AUC = 0.624$.

vi. Interpretation Caveats

- **Poverty coefficient flips sign** when conditioning on income and % White (unadjusted $OR = 0.98 \rightarrow$ adjusted $OR = 1.11$) — a classic suppression effect from collinearity, *not* evidence that poverty improves outcomes. The PCA SES index ($OR = 1.065$) is the cleanest single measure.
- **Inspection frequency is endogenous** to outcomes: failures trigger re-inspections, so conditioning on frequency risks collider bias on the SES coefficient. This remains the model’s most consequential limitation; future work should refit without frequency as a sensitivity check.
- **Business type explains more than SES** (Mobile food $OR \approx 1.42$; Retail food $OR \approx 1.18$), reinforcing that *what* an establishment does matters more than *where* it sits.

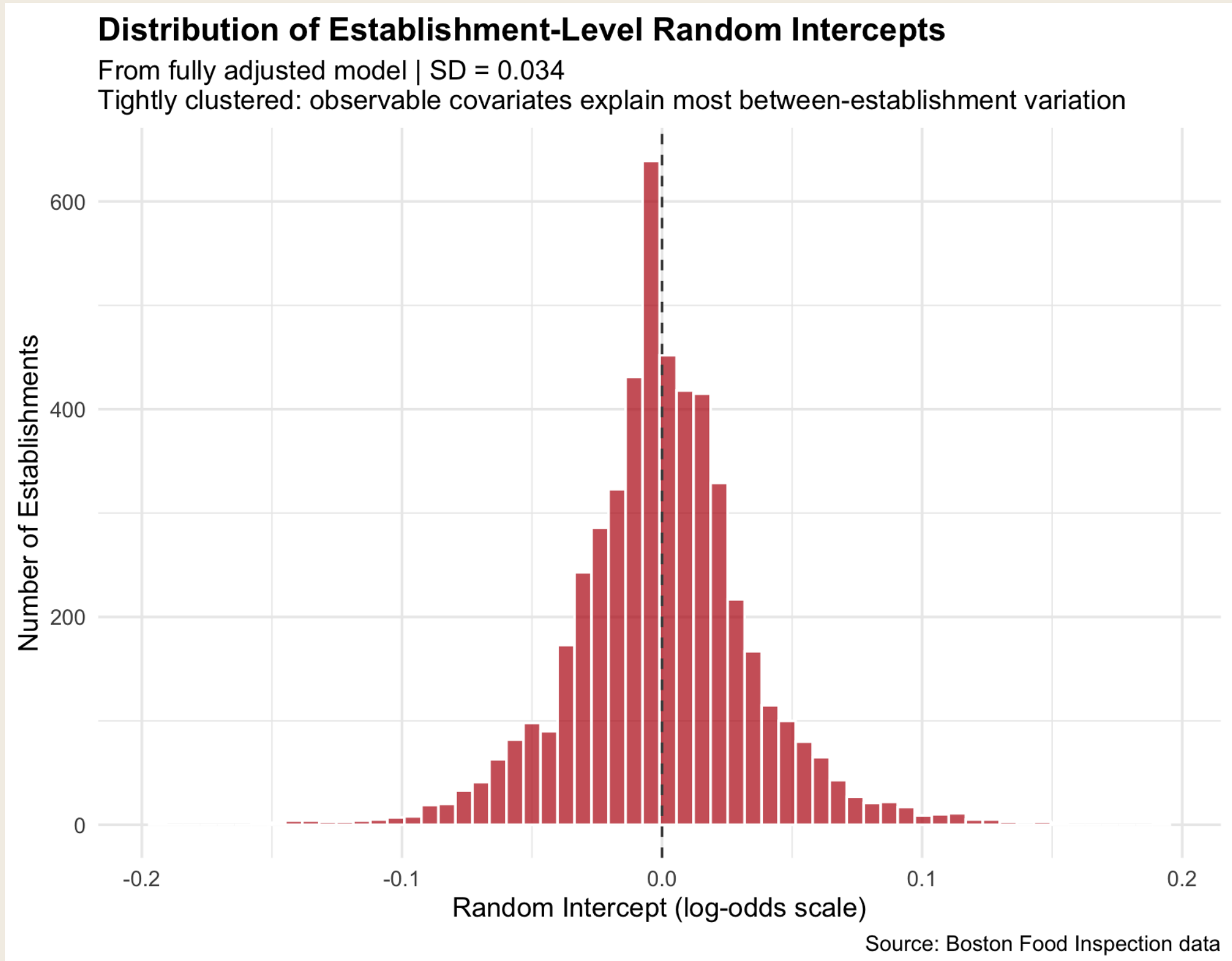


Figure 6. Establishment random intercepts cluster tightly around zero ($SD = 0.034$) once observable covariates are accounted for — confirming little residual heterogeneity.

vii. Conclusion

Neighbourhood income has a real but modest association with food inspection outcomes in Boston.

The effect is dwarfed by within-establishment variability and by differences across business types. The income gradient is statistically stable over the past decade.

These findings are consistent with a case for **establishment-level rather than area-based regulatory attention**, and push back against narratives that low-income Boston neighbourhoods systematically fail inspections at dramatically higher rates than wealthier ones. Whether establishment-level intervention would actually *improve* outcomes is a separate question this analysis cannot answer.